

■ 4.1 Programming Methodologies

■ 4.1.1 Approaches to Programming

There are several different approaches to “programming” in *Mathematica*. Which approach you choose in a particular case will depend on your computational background, and on the problems you are trying to solve.

All the approaches are in a sense ultimately equivalent, but one of them may be vastly more efficient for a particular problem, or may simply fit better with your way of thinking about the problem.

What this section does is to go through a number of different common programming methodologies, and explains how you can use them in *Mathematica*.

Procedural programming	specify step-by-step algorithms
Functional programming	specify a collection of functions to apply
Mathematical programming	specify mathematical relations

Three common approaches to *Mathematica* programming.